Palaeoclimate Stability and Modern Biodiversity

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1. Introduction

Currently, biodiversity varies in its distribution across the globe and is found in abundance in particular regions. The reason behind this could be attributed to what is known as the paleoclimate stability hypothesis where regions with climatic stability for longer periods of time could have offered habitats in which the species could flourish and diversify. Nonetheless, the correlation between climatic stability over longer periods of time and currently observed biodiversity can differ among geographical locations, taxa and different measures of biodiversity. Such knowledge could be useful in understanding the role played by climatic conditions historically in explaining current patterns of biodiversity. This literature review aims at investigating the following question: Do currently biodiversity rich regions have disproportionate paleoclimate stability? It analyzes works carried out at the global, regional and local geographical levels in their consideration of past climates and biodiversity and their interaction. The review highlights the differences among the conclusions reached and limitations in methodology used in these works, and identifies gaps in data which require further analysis based on existing data on past climates and biodiversity.

2. Review of Existing Studies

Research on how past climate shapes present day biodiversity now spans many taxa, diversity facets and ways of describing historical climate. The sixteen studies reviewed here are compared under five themes.

Defining and measuring past climate Climatic stability is not defined uniformly across these studies. It is represented by LGM to present anomalies, climate velocity, temporal variability, biome persistence and accessibility to glacial refugia. Rakotoarimivo et al. (2013) and Xu et al. (2019) used LGM to present anomalies but defined them in opposite directions, so the sign of reported effects must be interpreted according to each study's definition. Others used climate velocity (Ordonez and Svenning 2016; Feng et al. 2016; Acevedo and Sandel 2021). From 2018 onwards, some studies began using longer palaeoclimate sequences or multiple time slices: Blonder et al. (2018) summarised HadCM3 simulations from 0 to 120 ka as means and variability, and Scheiter et al. (2026) modelled biome stability across 2,605 time slices. Fastovich et al. (2024) tested 13 palaeoclimate estimates and found that relationships varied in strength and direction between them. This highlights the importance of testing palaeoclimate biodiversity relationships across multiple reconstructions rather than relying on a single one.

Taxa and Biodiversity Measures Plants have been considered in several papers, along with birds (Wang et al. 2020; Acevedo and Sandel 2021) and multi taxa communities (Huang et al. 2023; Oliveira et al. 2020; Magneville et al. 2025). Also, the studies consider functional diversity (Ordonez and Svenning 2015), phylogenetic composition (Xu et al. 2019), endemism (Feng et al. 2016), and beta diversity (Xu et al. 2023). For example, in woody plant species of China, palaeoclimate accounted for just 1.1% of richness variability but was more influential in affecting phylogenetic structure (Xu et al. 2019), and Huang et al.

Analytical approaches The following are some of the methods that the various researches adopted in order to establish the effect of historical climate on the biodiversity of an area. Spatial autoregression analysis was common due to the presence of spatial autocorrelation (Rakotoarimivo et al. 2013; Oliveira et al. 2020; Xu et al. 2023). In other cases, partial least square technique (Blonder et al. 2018) and random forest method (Acevedo and Sandel 2021;

Magneville et al. 2025) or structural equations were used in separating the direct climate effect from those via plant richness (Wang et al. 2020). Also, null models were used in verifying if the phylogenetic patterns were any different from the taxonomic ones (Xu et al. 2023); in the estimation of potential functional diversity (Ordonez and Svenning 2015) and testing for boundary bias in endemism (Feng et al. 2019).

Legacy vs. stability Blonder et al. (2018), Huang et al. (2023) and Xu et al. (2023) examine whether there is any legacy of past climate changes or variations for current functional composition or beta diversity, while Oliveira et al. (2020), Acevedo and Sandel (2021), Magneville et al. (2025) and Scheiter et al. (2026) examine stability directly. They find evidence of a signal in all cases except one. The association was found between stability and refugial accessibility on one hand, and functional diversity of European plants on the other (Ordonez and Svenning 2015, 2016); moreover, Magneville et al. (2025) identified past temperature stability as the main driver among five studied taxa. Past precipitation seems to matter more in Madagascar (Rakotoarimivo et al. 2013) and South America (Blonder et al. 2018), while past temperature is important for North America (Blonder et al. 2018) and China (Feng et al. 2016). However, there is no unanimous support; specifically, Scheiter et al. (2026) revealed low congruence between stable biomes and richness hotspots, and Fastovich et al.

Limitations Most studies are correlative, and several rely on a single LGM snapshot or coarse spatial resolution. Several also depend on incomplete trait data and imputation; about half the trait values were imputed in Ordonez and Svenning (2016). Rakotoarimivo et al. (2013) found that estimated relationships differed between palaeoclimate reconstructions, with the MIROC3.2 based estimate producing a weaker, negative effect than CCSM3. Costa et al. (2018) developed stability scenarios but did not directly test whether stable regions contain greater present day biodiversity, leaving the proposed consequences largely as hypotheses. Future work would benefit from ensembles, multiple time slices and clearer separation of stability from collinear modern drivers.

3. Critical Discussion and Research Gap

Across the sixteen studies, most found that past climate leaves some detectable mark on present day biodiversity, although how strong that mark is varies a lot. Stability or refugial accessibility was often linked to higher functional diversity, endemism or richness (Ordonez and Svenning 2016; Feng et al. 2016; Magneville et al. 2025). Where the studies disagree is on how much history matters once modern climate is accounted for. Scheiter et al. (2026) found weak overlap between stable biomes and richness hotspots. Xu et al. (2019) found that palaeoclimate effects on richness largely overlapped with current climate. Huang et al. (2023) showed that the direction of the effect depends on which diversity facet is measured.

Some limitations keep coming up. "Stability" is measured in different ways, so results are hard to compare directly, and many studies rely on a single LGM snapshot, coarse grids or imputed traits. Geographically, the studies cluster in China, Europe and North America, with little coverage of Africa beyond Madagascar. Some also use temperature alone (Acevedo and Sandel 2021). Future work could combine several palaeoclimate reconstructions and time slices, include precipitation, and test stability directly against functional and phylogenetic diversity in understudied regions.

4. Conclusion

Overall, the reviewed studies suggest that past climate has left a detectable, though often modest, imprint on present day biodiversity. Stable climates and access to glacial refugia were most often linked to higher functional diversity, endemism or richness, but the strength of these links depended on the taxon, region and diversity facet examined. Most studies were correlative, typically relating LGM based climate measures to current diversity patterns using spatial regression or machine learning models. The main limitation is that stability is defined and measured inconsistently, and many studies rely on a single palaeoclimate reconstruction. Future research should test stability more directly, using multiple reconstructions and time slices, and extend coverage to under studied regions.

5. References

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